

Design a low-power **64-bit RISC-V compute subsystem** for **mobile XR** under a **3 W, 3 nm-class LP envelope**, returning a **DSE report with evidence and rejected alternatives**.

Intent

Improve mobile XR efficiency under power, reliability, memory, and deployment constraints.

Task

Bounded DSE for a RISC-V compute subsystem; start with an accelerator and memory slice.

Representation /world model

XRBench traces, architecture parameters, compiler assumptions, power model, constraints.

Environment /tool interface

Simulator or cost model plus workload harness; actions change tiling, memory, or vector width.

Method role

Generate candidates, predict cost, optimize, critique assumptions, and write evidence reports.

Feedback budget

Many cheap proxies, fewer simulations, rare high-fidelity checks before review.

Evidence

Pareto comparison over latency, energy, area proxy, memory traffic, QoS, and sensitivity.

Negative traces

Violates 3 W, misses real-time, exceeds bandwidth, unsupported software or fidelity failure.

Rejection authority

Checks, tool failures, QoS misses, review.

Human decision

Refine, escalate fidelity, change scope, or reject.

Example card, not a completed design claim.